

Autonomous Orbital Bioscience Infrastructure

- A free-flying CubeSat BioLab platform for autonomous antimicrobial drug-response testing in orbit.
- A nurse-led bioscience research satellite project in low Earth orbit for advancing healthcare.
- Autonomous orbital wet-lab infrastructure.
- Crew-free biological experimentation platform.
- Repeatable microgravity bioscience missions.
- Scalable orbital research and development architecture.
- Designed for translational health applications.

Orbital Bioscience Needs a Scalable Operating Model

- International Space Station access remains limited and expensive.
- Human-dependent workflows reduce scalability.
- Mission integration overhead slows experimentation.
- Commercial orbital bioscience infrastructure remains underdeveloped.
- Autonomous data downlink with future sample-return capability.

The Problem

Most Orbital Life Science Missions Still Depend on High-Touch Operations

- Crew dependence.
- International Space Station scheduling bottlenecks.
- Custom payload integration.
- High operational overhead.
- Microgravity alters microbial behavior and drug response.

A Free-Flying Orbital Wet-Lab Platform

- Sealed microfluidics.
- Timed dosing.
- Environmental sensing.
- Autonomous execution.
- Onboard data logging.
- Reduced operational burden and lower mission cost.

Software-Defined Architecture Changes the Economics of Orbital Bioscience

- Experiment design and payload integration.
- Launch and deployment.
- Crew-free orbital operation.
- Autonomous data downlink and analysis.
- Repeatable mission workflows.

Enabling Practical Orbital Bioscience

- Microbial adaptation research.
- Drug-response behavior studies.
- Pharmaceutical pilot studies:
- Translational biologic research.
- Contract orbital research and development services.

Nurse-Led Leadership at the Frontier of Aerospace Bioscience

- Dr. Joannes Paulus Tolentino Hernandez.
- Master of Science in Aerospace Physiology candidate.
- NASA Twins Study contributor.
- Research Fellow, Advanced SpaceLife Research Institute.
- Principal investigator and co-principal investigator on multiple spaceflight initiatives.

From Pilot Missions to Repeatable Orbital Research Infrastructure

- Standardized assay packages.
- Hosted payload integration:
 - Contract orbital research and development services.
- Recurring flight campaigns.
- Built for modularity and designed for scale.

A Strategic Entry Point into Orbital Bioscience

- Investment needed for the current project phase: \$250K.
- Payload prototyping.
- Biosafety systems development.
- Environmental qualification testing.
- Commercialization preparation.

Building the Infrastructure Layer for Orbital Bioscience

- Modular autonomous spacecraft infrastructure.
- Reduces mission cost.
- Accelerates scientific iteration.
- Supports commercial orbital bioscience ecosystems.
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